A Real-Time Research Project Report

on

IOT VEHICLE ACCIDENT ALERT SYSTEM

submitted in partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

in

COMPUTER SCIENCE AND ENGINEERING (AI & ML)

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI&ML)

CERTIFICATE

This is to certify that the Real-Time Research Project Report entitled **"IOT VEHICLE ACCIDENT ALERT SYSTEM"** being submitted by **K,Nagarani [247R1A6685] , Abdul Hameed[247R1A6666] , E.Nithya Shree [257R5A6678] , Sheema Begum [257R5A6606]** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING (AI & ML)** to the **Jawaharlal Nehru Technological University, Hyderabad** is a record of bonafide work carried out by them under my guidance and supervision during the Academic Year 2025 – 26.

The results embodied in this report have not been submitted to any other University or Institute for the award of any other degree.

A.Prashanthi
Assistant Professor
Project Guide

Dr. S Rao Chintalapudi
Head of the Department

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ABSTRACT

Road accidents are a leading cause of fatalities worldwide, often resulting in severe injuries or loss of life due to delays in medical assistance. In many cases, victims are unable to call for help immediately due to unconsciousness or severe trauma.

The time immediately following an accident, known as the **"Golden Hour,"** is critical for saving lives and improve survival chances.The **Accident Alert System** is an IoT-based safety solution designed to significantly reduce the time between a vehicle accident and the arrival of emergency services called by registered contacts

Using gyrosensor(which identifies the motion of the vehicle)which detects the accident and send the signal to ESP8266 microcontroller and this microcontroller send signal to L298N MOTOR DRIVER IC which controls the DC Motor then the alert pop up will come to registered contacts with the help of ESP8266 Microcontroller.

LIST OF FIGURES

S. No	Figure Name	Page No
Figure 4.2	System Design	6
Figure 4.3.1	Class Diagram	7
Figure 4.3.2	Use Case Diagram	8
Figure 4.3.3	Sequence Diagram	9
Figure 4.3.4	Component Diagram	9
Figure 4.3.5	Activity Diagram	11
Figure 5.2	Installation Steps	13

Table of Contents

Chapter Title	Page No
Abstract	i
List of Figures	ii
1.Introduction	1
1.1 Motivation	1
1.2 Problem Statement	2
1.3 Project Objective	3
2. Literature Survey	3
2.1 IOT – Based Accident Detection Systems	3
2.2 Sensor-Based Accident Detection Techniques	3
2.3 Role of Microcontrollers and ESP8266	3
2.4 Communication and Alert Mechanisms	3
2.5 Challenges in Existing Systems	4
2.6 Need for Proposed System	4
2.7 Summary	4
3. Requirement Specification	
3.1 Software Requirements	5
3.2 Hardware Requirements	5
3.3 Functional Requirements	5
3.4 Non-Functional Requirements	6
4. System Design	
4.1 System Design	7
4.2 Proposed Methodology	7
4.3 UML Design	8
4.3.1 Class Diagram	8
4.3.2 Use Case Diagram	9
4.3.3 Sequence Diagram	10
4.3.4 Component Diagram	11
4.3.5 Activity Diagram	12
5. Implementation	
5.1 Hardware Implementation	13
5.2 Software Implementation	14
6. Testing	
6.1 Introduction to Testing	18
6.2 Test Cases	19
7.Results	23
8.Conclusion	24
9.Future Scope	25
10.References	

1. INTRODUCTION

1.1 Motivation

With the rapid increase in the number of vehicles, road accidents have become a major concern worldwide. One of the biggest challenges after an accident occurs is the delay in providing timely medical assistance, which can lead to severe injuries or even loss of life. In many situations, victims are unable to seek help due to unconsciousness or lack of immediate support. To address this issue, an automated and intelligent accident detection system is required. This project presents an IoT-based Accident Alert System that uses a gyrosensor to detect sudden motion or impact, indicating a possible accident. The detected data is processed by the ESP8266 microcontroller, which plays a crucial role in analyzing the situation and triggering necessary actions.

Once an accident is detected, the system automatically sends alert messages to pre-registered contacts, ensuring quick response and assistance. Additionally, the microcontroller communicates with the L298N Motor Driver IC to control a DC motor, which can be used for safety mechanisms or system responses.

This system aims to reduce response time, improve emergency communication, and enhance overall road safety using smart and connected technology.

1.2 Problem Statement

Road accidents often lead to serious injuries or fatalities due to delays in detecting the accident and informing emergency contacts. In many cases, victims are unable to call for help immediately, which increases the risk of loss of life. There is a need for an automated system that can quickly detect accidents and send alerts without human intervention.

The existing systems lack real-time monitoring, immediate notification, and integrated control mechanisms. Therefore, a reliable solution is required that can detect sudden motion or impact using sensors, process the data efficiently, and automatically notify registered contacts.

This project aims to develop an IoT-based accident alert system using a gyrosensor and ESP8266 microcontroller that detects accidents, processes the data, sends emergency alerts, and also controls a DC motor through an L298N motor driver for additional safety or response actions.

1.3 Project Objective

- Detect accidents using a gyrosensor
- Process data using ESP8266
- Send automatic alerts to contacts
- Reduce emergency response time
- Control DC motor using L298N driver
- Provide a low-cost IoT safety solution

2. LITERATURE SURVEY

The development of IoT-based accident detection and alert systems has gained significant attention due to the increasing number of road accidents and the need for faster emergency response.

2.1. IoT-Based Accident Detection Systems

Several research works have focused on designing accident detection systems using IoT technologies. A study by Simran et al. (2022) highlights that road accidents cause approximately 1.3 million deaths annually, mainly due to delays in providing medical assistance. Their system uses sensors, GPS, and communication modules to automatically alert family members and emergency services.

Another research work emphasizes that IoT systems can continuously monitor vehicle conditions and send real-time alerts using wireless communication technologies like GSM, Wi-Fi, and cloud platforms.

2.2. Sensor-Based Accident Detection Techniques

Most existing systems use sensors such as accelerometers and gyroscopes to detect sudden changes in motion or impact. These sensors measure parameters like acceleration, tilt, and vibration to determine whether an accident has occurred.

For example, ADXL345 accelerometer-based systems detect abrupt changes in motion and send alerts to predefined contacts.

2.3. Role of Microcontrollers and ESP8266

Microcontrollers play a crucial role in processing sensor data and triggering alerts. Many research works use Arduino, Raspberry Pi, or ESP8266 for real-time processing.

ESP8266 is widely used due to its built-in Wi-Fi capability, enabling efficient communication with cloud servers and mobile devices. It processes sensor data and sends notifications instantly when abnormal conditions are detected.

2.4. Communication and Alert Mechanisms

Communication technologies such as GSM, GPS, and Wi-Fi are commonly used to send accident alerts.

GPS helps in identifying the exact location of the accident

GSM/Wi-Fi sends alerts to Registered contacts

Cloud platforms store and process real-time data

Research shows that automated alert systems significantly reduce response time and improve survival rates.

2.5. Challenges in Existing Systems

Despite advancements, current systems face several limitations:

False alarms due to sensor inaccuracies

Limited real-time processing in low-cost systems

Lack of integrated control mechanisms

Studies suggest that improving sensor accuracy and combining multiple technologies can enhance system reliability.

2.6. Need for Proposed System

From the literature, it is clear that although many accident detection systems exist, there is still a need for:

Low-cost and efficient solutions

Real-time accident detection using gyrosensors

Fast and automatic alert mechanisms

Integration of additional control systems (like motor control)

2.7 Summary

The reviewed literature indicates that IoT-based accident detection systems are effective in improving road safety. However, existing systems still have limitations in terms of accuracy, cost, and integration.

The proposed system using a gyrosensor, ESP8266 microcontroller, and L298N motor driver aims to overcome these issues by providing a reliable, real-time, and automated accident detection and alert solution.

3. REQUIREMENT SPECIFICATION

3.1 Software Requirements

- Arduino IDE
- Embedded C / Arduino Programming
- IoT platform (Blynk)

3.2 Hardware Requirements

- ESP8266 Microcontroller
- Gyrosensor (e.g., MPU6050)
- L298N Motor Driver
- DC Motor
- Power Supply
- Buzzer

3.3 Functional Requirements

- Detect accident using gyrosensor
- Monitor vehicle movement continuously
- Process data using ESP8266
- Send alert message to contacts
- Work automatically without user help
- Control DC motor using L298N3 MotorDriver

3.4Non-Functional Requirements

- Should be accurate (less false alarms)
- Should respond quickly
- Should be low cost
- Should use less power
- Should be easy to use
- Should be easy to maintain
- Should work with Wi-Fi properly

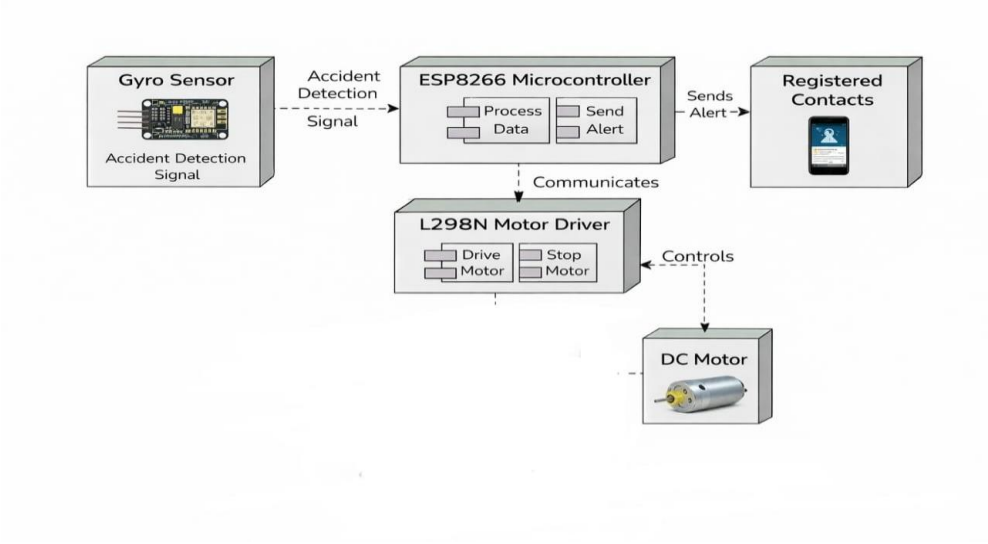
4. SYSTEM DESIGN

4.1 System Design

The system consists of:

- Gyrosensor (detects motion/impact)
- ESP8266 Microcontroller (processes data & sends alerts)
- L298N Motor Driver (controls motor)
- DC Motor (performs action)
- Power Supply
- Wi-Fi Network

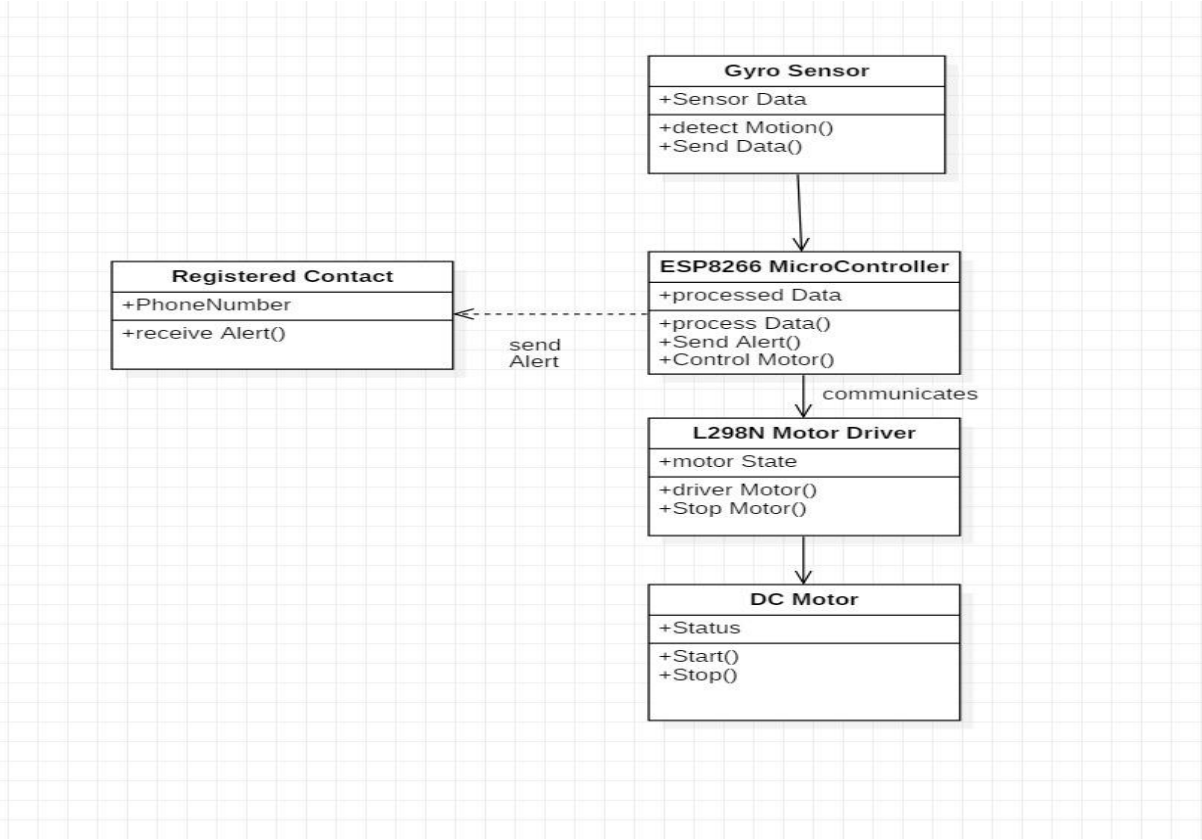
4.2 Proposed Methodology



1. The gyrosensor continuously monitors the vehicle movement.
2. If sudden motion or impact is detected, it sends a signal to the ESP8266.
3. The ESP8266 processes the data and checks if it is an accident.
4. If an accident is confirmed:
    - Alert message is sent to registered contacts via Wi-Fi
    - Motor is controlled using L298N driver
    - The system works automatically without human intervention.

4.3 UML DESIGN

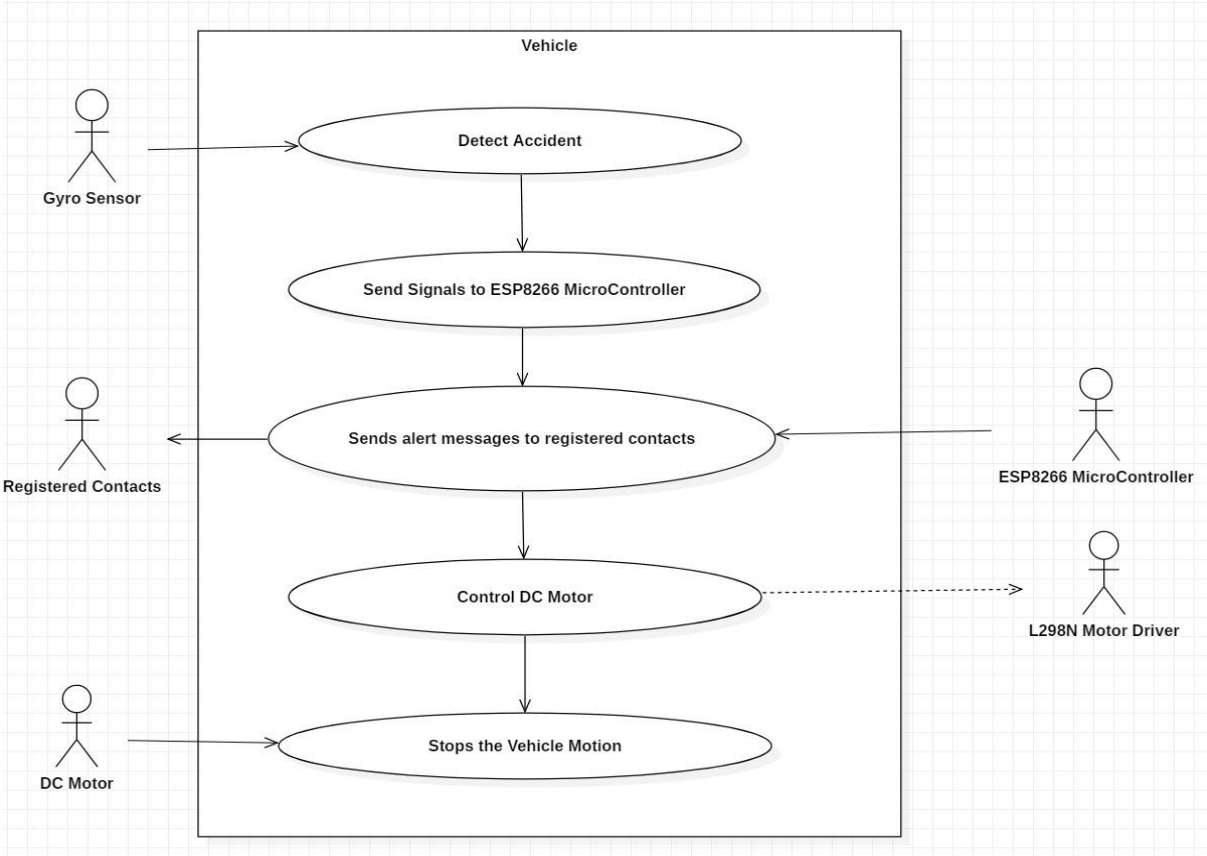
4.3.1 Class Diagram



The system consists of five main classes:

- Gyro Sensor
- ESP8266 Microcontroller
- L298N Motor Driver
- DC Motor
- Registered Contact

4.3.2 Use Case Diagram



1. Detect Accident

   Gyro sensor identifies abnormal motion
2. Send Signals to ESP8266

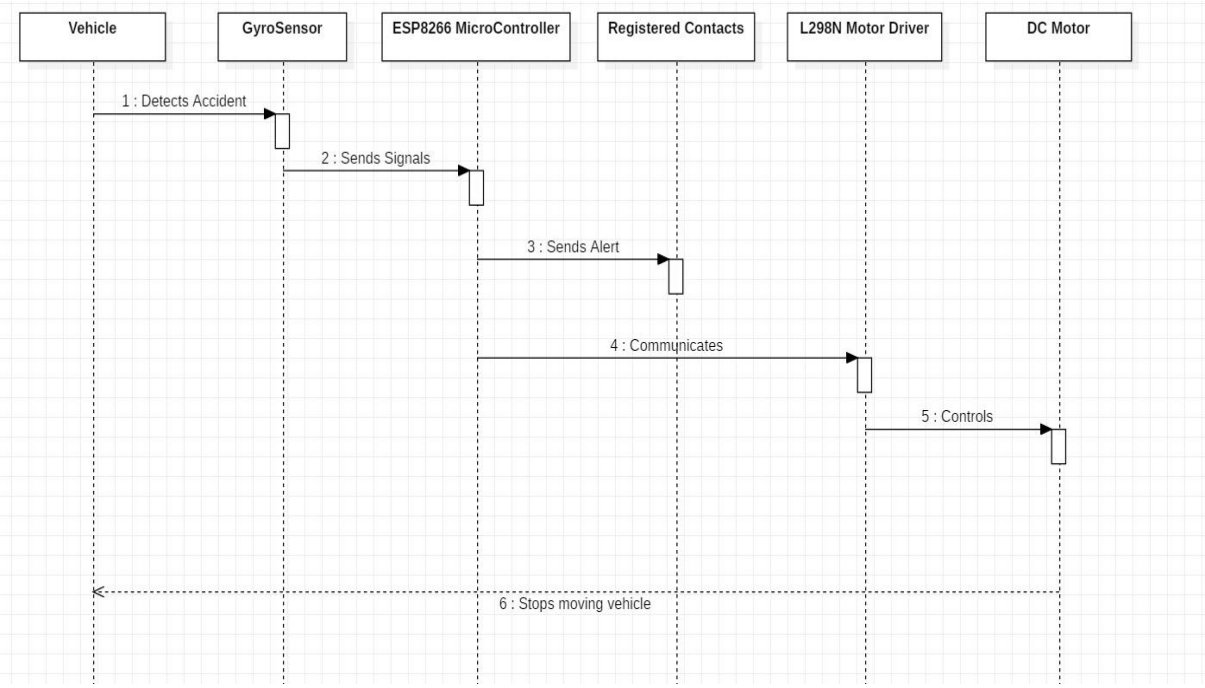
   Sensor sends data to the microcontroller
3. Send Alert Messages

   ESP8266 sends notifications to registered contacts
4. Control DC Motor

   Microcontroller sends control signals to motor driver
5. Stop Vehicle Motion

   Motor stops the vehicle for safety

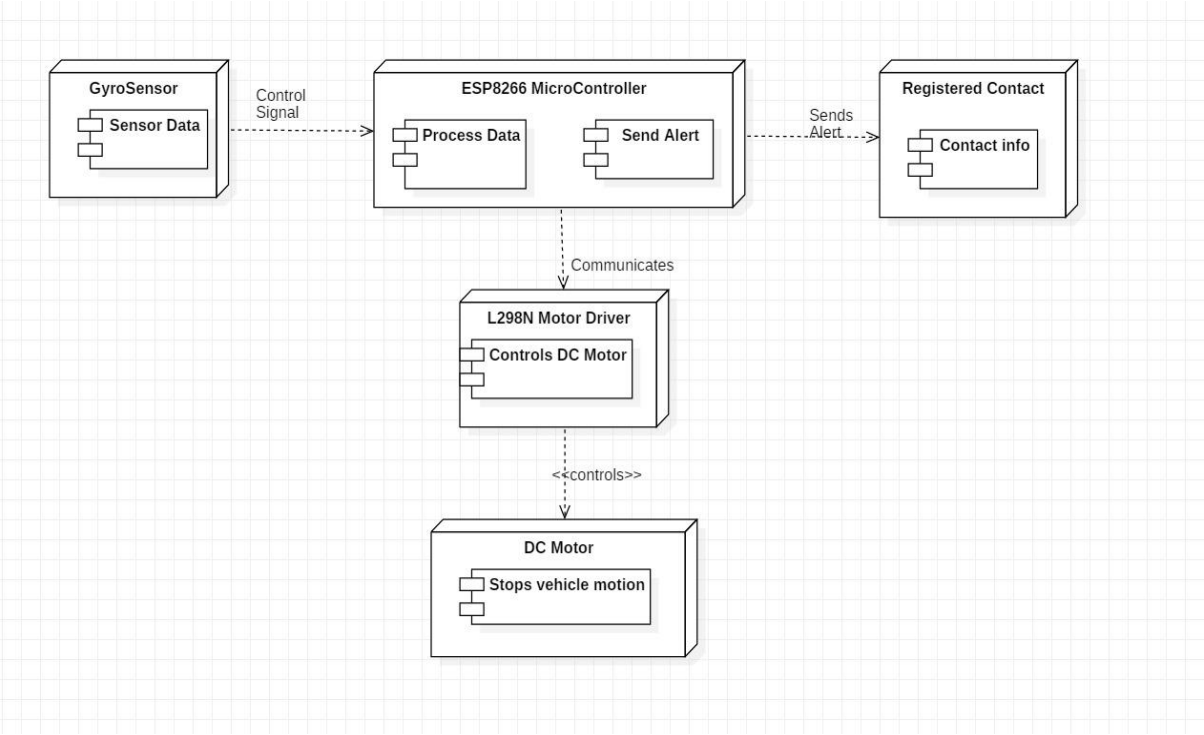
4.3.3 Sequence Diagram



Flow:

Detect → Send Data → Process → Alert → Control Motor → Stop Vehicle

4.3.4 Component Diagram



Main Components

- Gyro Sensor

  → Collects motion data and detects accidents
- ESP8266 Microcontroller

  → Processes data and controls the system

  → Sends alert messages
- Registered Contact

  → Receives alert notifications
- L298N Motor Driver

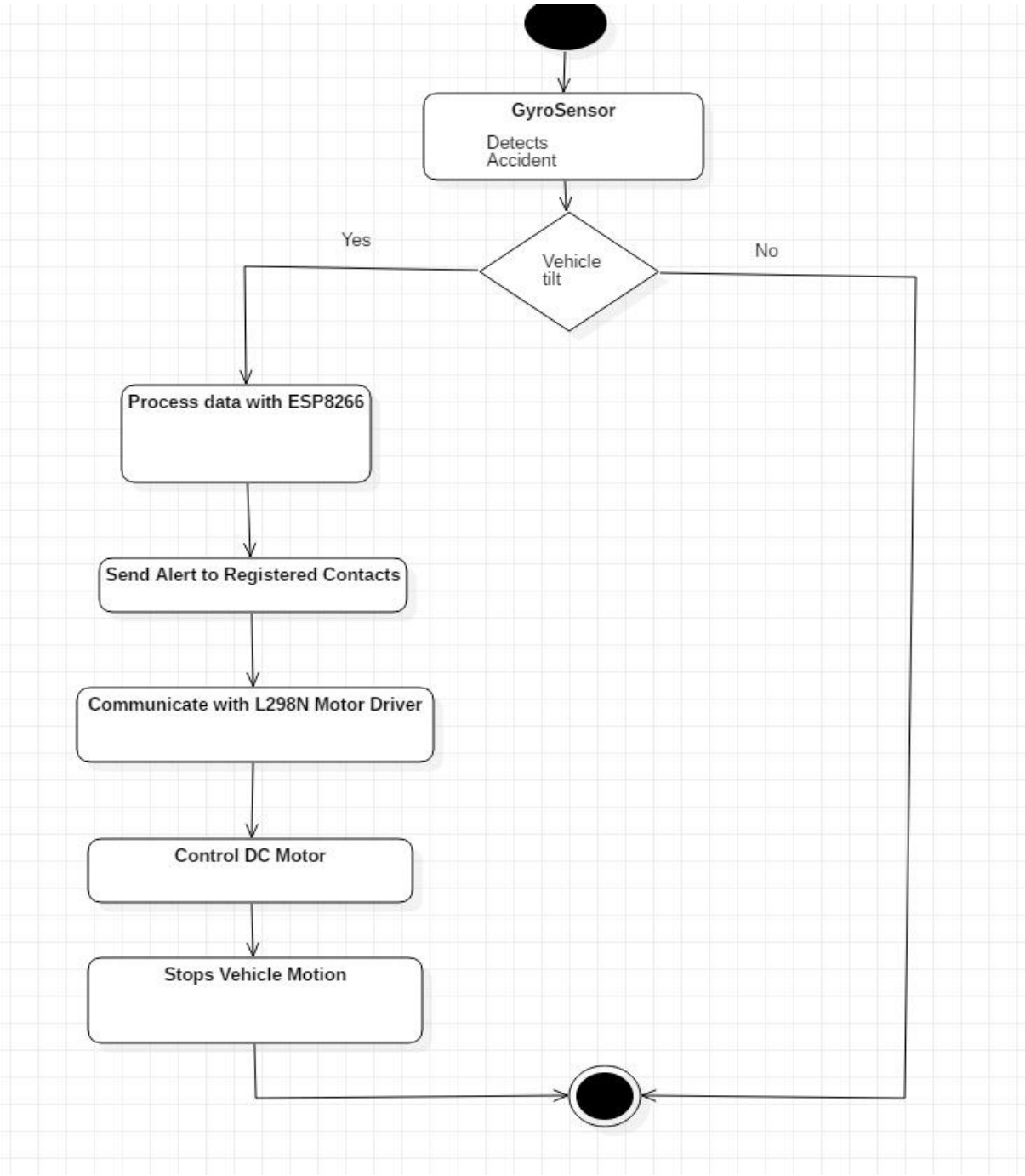
  → Acts as an interface between controller and motor
- DC Motor

  → Stops vehicle movement

4.3.5 Activity Diagram

Activities Involved

- Start → System begins monitoring
- Gyro Sensor Detects Accident → Checks vehicle motion
- Decision: Vehicle Tilt?
  - Yes → Accident detected
  - No → No action, system ends
- Process Data (ESP8266) → Analyzes sensor data
- Send Alert to Contacts → Notifies registered users
- Communicate with Motor Driver → Sends control signal
- Control DC Motor → Activates motor action
- Stop Vehicle Motion → Stops the vehicle
- End → Process completes


GyroSensor
Detects Accident
Vehicle tilt
Yes
No
Process data with ESP8266
Send Alert to Registered Contacts
Communicate with L298N Motor Driver
Control DC Motor
Stops Vehicle Motion

5. IMPLEMENTATION

5.1 Hardware Implementation

1.ESP8266 Microcontroller

This is the main brain of the circuit.

It contains:

ESP8266 Wi-Fi chip , USB-to-serial converter

Functions:

Executes the program (code).

Sends control signals to the motor driver.

Provides wireless communication (Wi-Fi).

Operates at 3.3V logic level

2.Gyrosensor

Besides sensing the angular velocity, Gyroscope sensors can also measure the motion of the object. For more robust and accurate motion sensing, in consumer electronics Gyroscope sensors are combined with Accelerometer sensors.

3.L298N Motor Driver

This L298N Motor Driver Module is a high power motor driver module for driving DC and Stepper Motors.This module consists of an L298 motor driver IC and a 78M05 5V regulator.L298N Module can control up to 4 DC motors, or 2DC motors with directional and speed control.

4.Buzzer

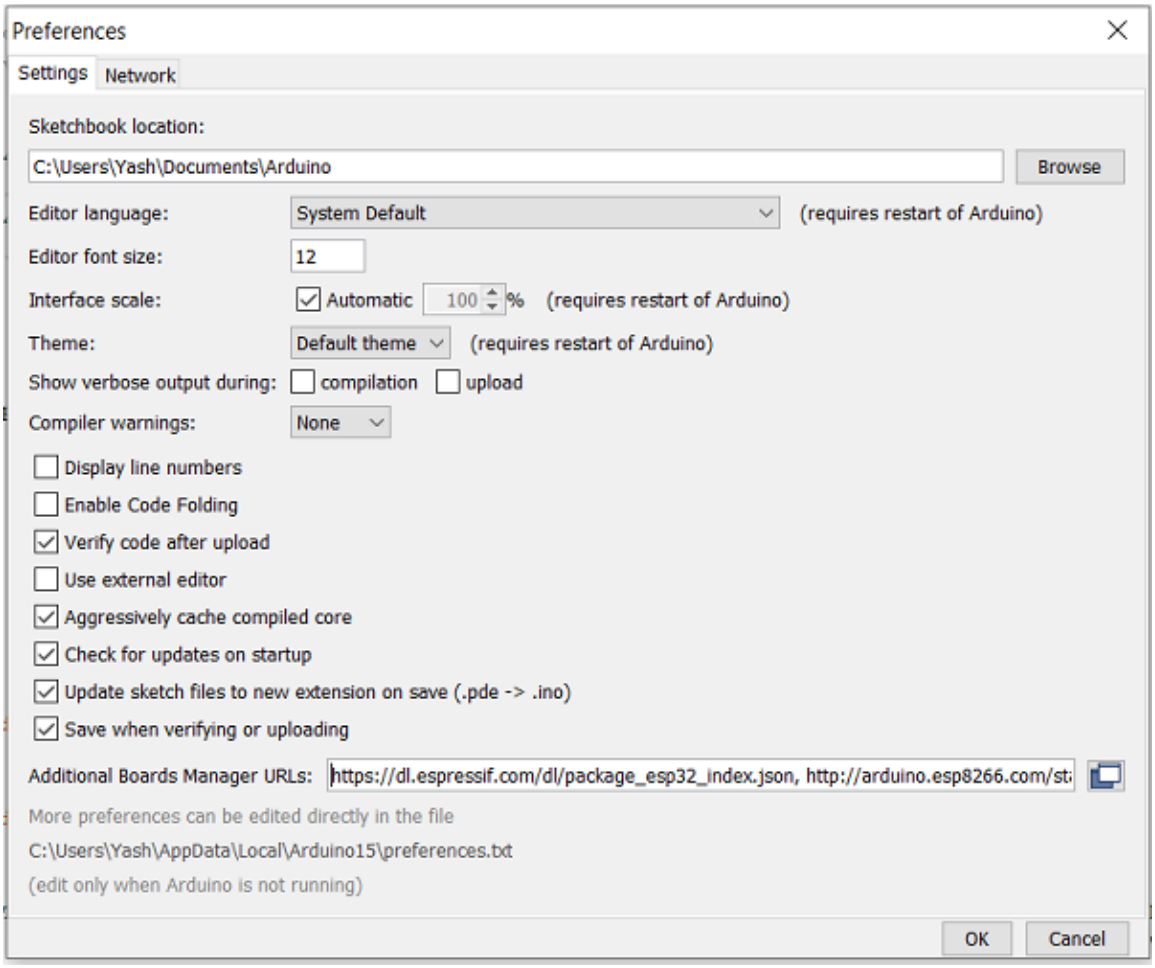
An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the generated signals to sound. Generally, it is powered through DC voltage and used in alarm devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren.

5.2 Software Implementation

1.ESP8266 Wi-Fi module

ESP8266 comes up with powerful processing speed on board. Storage space of this module is also high allowing it to integrate with other devices like sensors. To make this module compatible with other development boards, we have to do level shifting of voltages externally because this board doesn't come up with on-board voltage regulator.

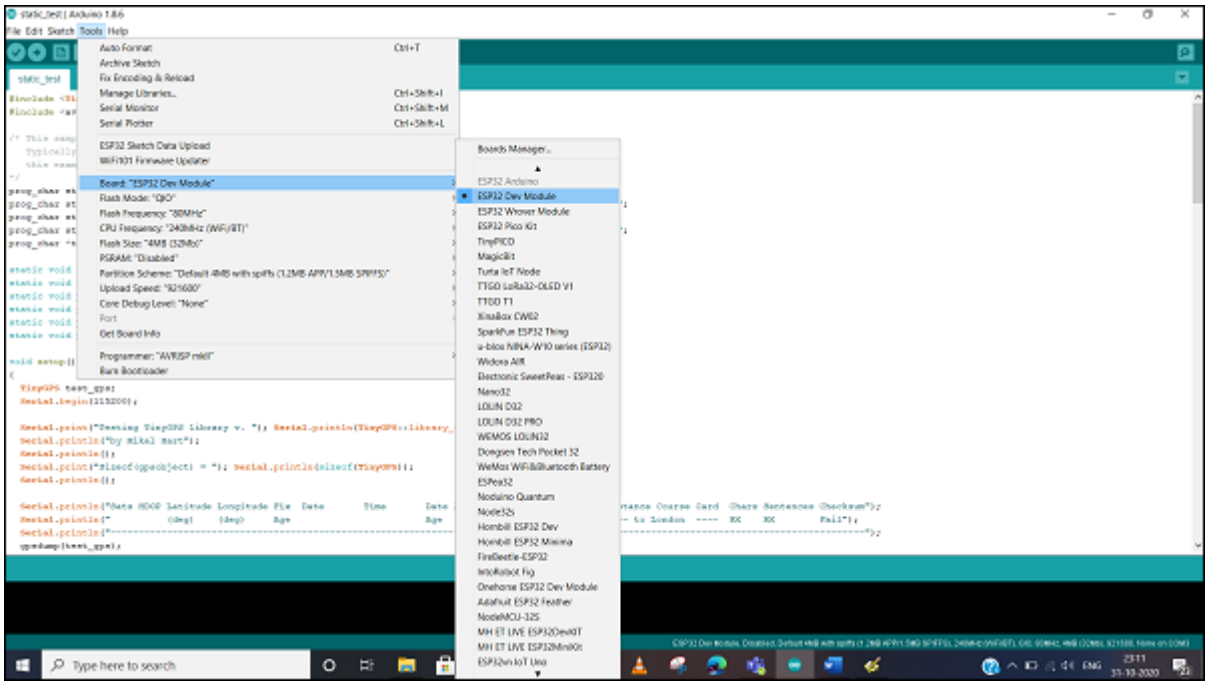
Installation Steps



Go to Tools –> Board–> Boards Manager. A pop–up would open up. Search for ESP8266and install the ESP8266by Espressif Systems board. The image below shows the board already installed because I had installed the board before preparing this tutorial.

Verifying the Installation

Once your ESP8266board has been installed, you can verify the installation by going to Tools –> Boards. You can see a whole bunch of boards under the ESP8266**Arduino** section. Choose the board of your choice. If you are not sure which board best represents the one you have, you can choose ESP8266**Module**



2.ARDUINO PROGRAMMING STRUCTURE

we will learn more new terminologies used in the Arduino world. The Arduino software is open- source. The source code for the Java environment In this chapter, we will study in depth, the Arduino program structure is released under the GPL and the C/C++ microcontroller libraries are under the LGPL.

Loop( ) function

Void setup ( )

{

}

PURPOSE:

The setup() function is called when a sketch starts. Use it to initialize the variables, pin modes, start using libraries, etc. The setup function will only run once, after each power up or reset of the Arduinoboard.

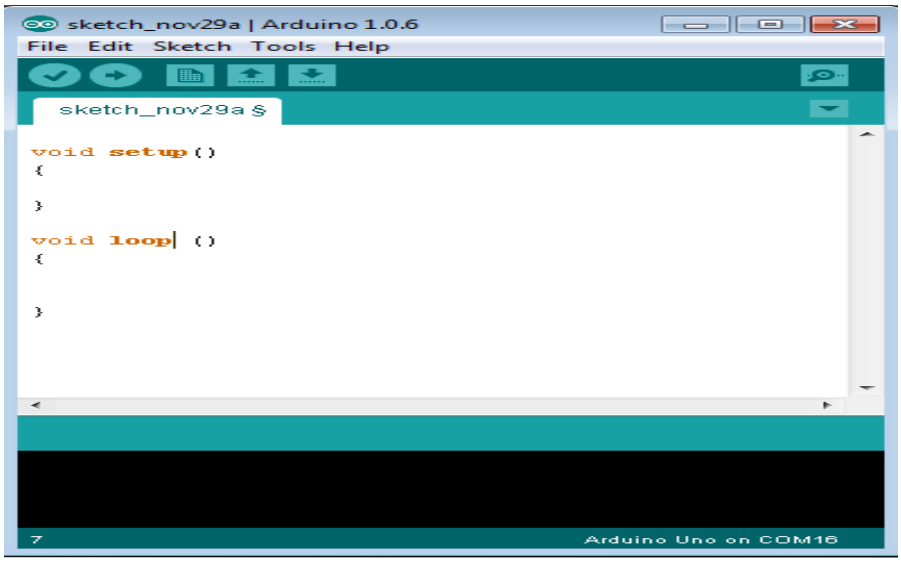
PURPOSE:

After creating a setup() function, which initializes and sets the initial values, the loop() function does precisely what its name suggests, and loops secutively, allowing your program to change and respond. Use it to actively control the Arduinoboard.

INPUT

OUTPUT

RETURN



6. TESTING

6.1 Introduction to Testing

The system was tested to ensure proper functioning of all components and accurate accident detection.

Sensor Testing:

The gyrosensor was tested by creating sudden movements and impacts. It successfully detected abnormal motion.

Microcontroller Testing:

The ESP8266 was tested to verify correct processing of sensor data and decision-making for accident detection.

Alert System Testing:

Alerts were tested by simulating accidents. Messages were successfully sent to registered contacts via Wi-Fi.

Motor Driver Testing:

The L298N motor driver was tested to ensure proper control of the DC motor during system activation.

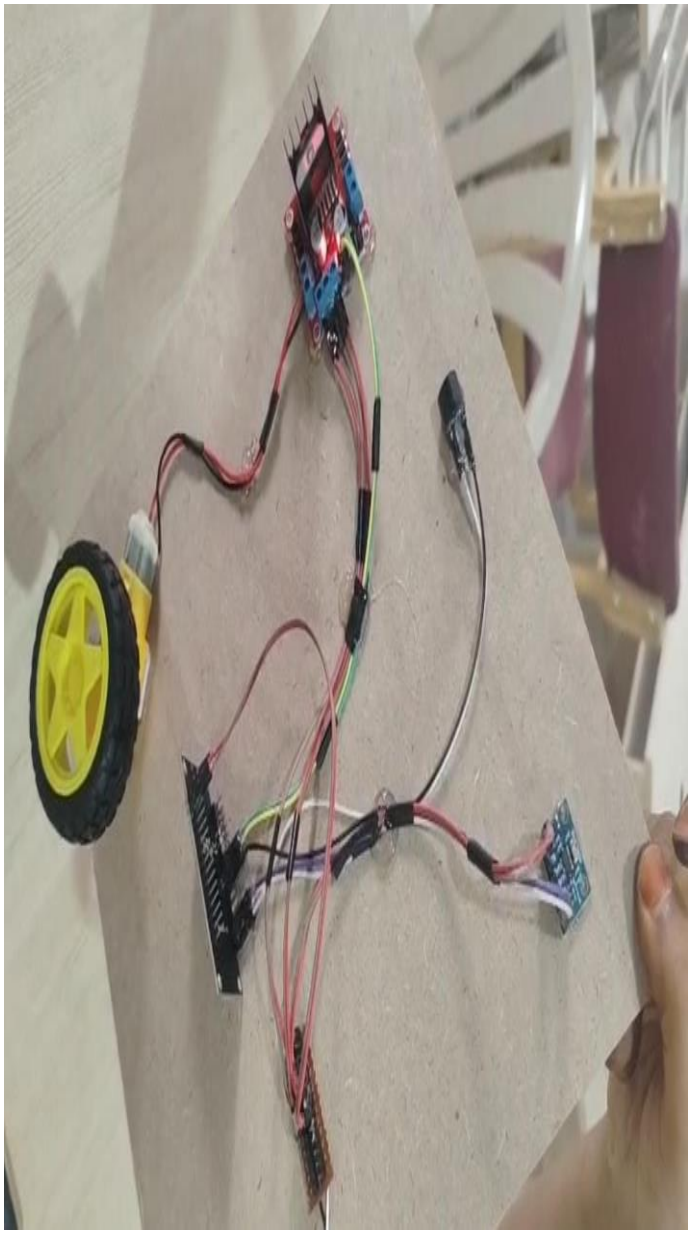
Overall System Testing:

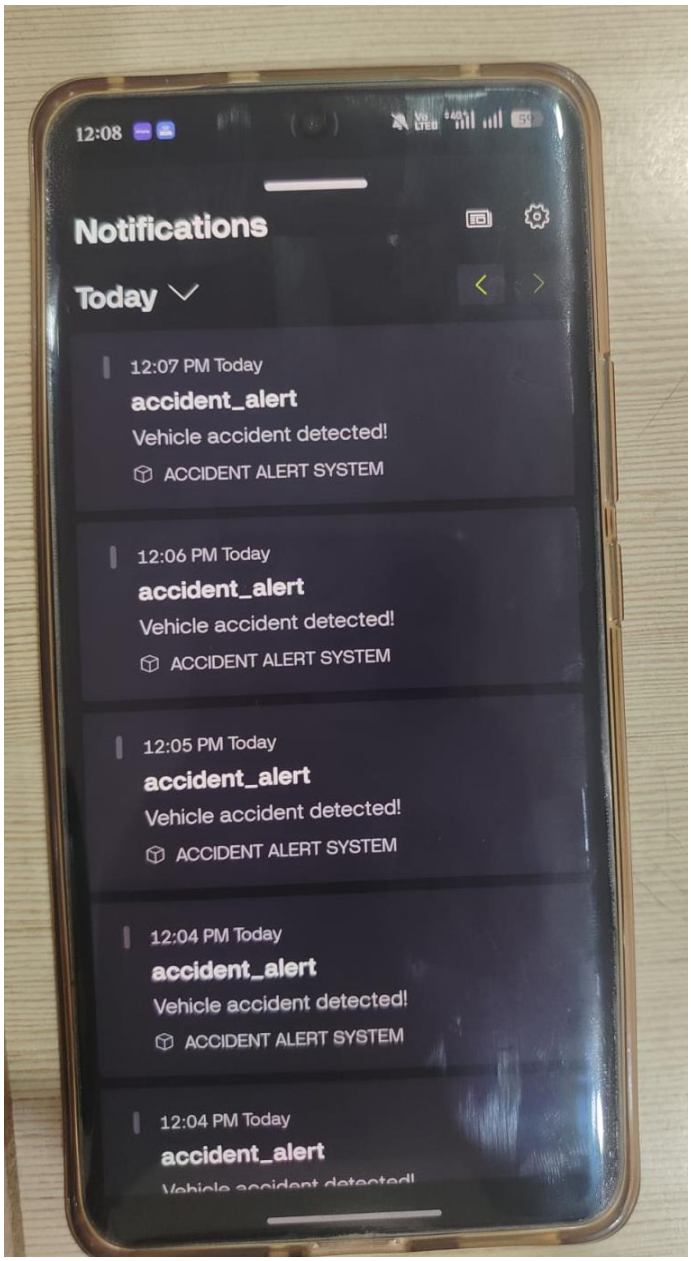
The complete system was tested by integrating all components. It worked correctly by detecting accidents, sending alerts, and controlling the motor automatically.

Test Cases Table

Test Case ID	Input/Condition	Result
TC01	Sudden movement/impact	Pass
TC02	No movement	Pass
TC03	High tilt/impact value	Pass
TC04	Sensor data input	Pass
TC05	Accident detected	Pass
TC06	Network available	Pass
TC07	Accident detected	Pass
TC08	All components connected	Pass

7. RESULTS




12:08
Notifications
Today
12:07 PM Today
accident_alert
Vehicle accident detected!
ACCIDENT ALERT SYSTEM
12:06 PM Today
accident_alert
Vehicle accident detected!
ACCIDENT ALERT SYSTEM
12:05 PM Today
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Vehicle accident detected!
ACCIDENT ALERT SYSTEM
12:04 PM Today
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Vehicle accident detected!
ACCIDENT ALERT SYSTEM
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accident_alert

8. CONCLUSION

The Accident Alert System presented in this project successfully demonstrates the use of Internet of Things (IoT) technology to enhance road safety and reduce accident-related fatalities. By automatically detecting vehicle motion abnormalities using a gyrosensor, the system is able to identify accident conditions without relying on human intervention.

The detected signal is processed by the ESP8266 microcontroller, which coordinates system actions and communicates with the L298N motor driver to control vehicle movement when required.This automated approach ensures that timely alerts can be generated even when the victim is unconscious or unable to seek help manually.

By addressing the critical “Golden Hour,” the proposed system significantly reduces the delay in emergency response, thereby increasing the chances of survival and minimizing the severity of injuries. The system is cost-effective, reliable, and suitable for real-time implementation in vehicles.

9. FUTURE SCOPE

- Integration with GPS for location tracking
- Direct ambulance notification system
- AI-based accident prediction
- Mobile app for monitoring
- Cloud data analytics
- Integration with smart city systems

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